

Supplementary Information: Electro-optic non-reciprocal polarization rotation in lithium niobate

Oğulcan E. Örsel ¹, and Gaurav Bahl ²

¹ Department of Electrical & Computer Engineering,

² Department of Mechanical Science and Engineering,
University of Illinois at Urbana–Champaign, Urbana, IL 61801 USA

S1 Calculation of the polarization rotation rate

In general, any three-wave mixing process that leverages a perturbation of the dielectric permittivity must satisfy the phase matching requirements:

$$\hbar\omega_2 = \hbar\omega_1 \pm \hbar\Omega \quad (\text{S1})$$

$$\hbar\vec{k}_2 = \hbar\vec{k}_1 \pm \hbar\vec{q} \quad (\text{S2})$$

where ω_1 and ω_2 represent the optical frequencies, Ω represents the frequency of the perturbing field, $\vec{k}_1$ and $\vec{k}_2$ represent the momentum of the photons, and $\vec{q}$ represents the momentum of the perturbing field. In order to simplify the analysis, suitable for our experimental case, we consider optical modes that are propagating along a 1D waveguide in the z-direction as:

$$\mathbf{E}^{\text{TM}}(\mathbf{r}) = a_1(z) \mathcal{E}^{\text{TM}}(\mathbf{r}_\perp) e^{i(k_1 z - \omega_1 t)} + c.c. \quad (\text{S3})$$

$$\mathbf{E}^{\text{TE}}(\mathbf{r}) = a_2(z) \mathcal{E}^{\text{TE}}(\mathbf{r}_\perp) e^{i(k_2 z - \omega_2 t)} + c.c. \quad (\text{S4})$$

Here $\mathbf{r}$ is the 3D position vector and $\mathbf{r}_\perp$ is the position vector in the cross-sectional plane transverse to the propagation direction along z, and we separate the transverse modal distributions ($\mathcal{E}^{\text{TE}}(\mathbf{r}_\perp)$ and $\mathcal{E}^{\text{TM}}(\mathbf{r}_\perp)$) from the modal amplitudes ($a_1(z)$ and $a_2(z)$). Since we are interested in the electro-optic effect we can write the material perturbation as an RF input field:

$$\mathbf{E}^{\text{RF}}(\mathbf{r}) = b(z) \mathcal{E}^{\text{RF}}(\mathbf{r}_\perp) e^{i(qz - \Omega t)} + c.c. \quad (\text{S5})$$

We can then assume that the dynamic refractive index grating (due to electro-optic modulation) created by this RF wave can be expressed under slowly varying wave approximation as:

$$\Delta n_{\text{EO}} = \Delta n_{\text{EO}} e^{i(qz - \Omega t)} + c.c. \quad (\text{S6})$$

where $\Delta \mathbf{n}_{\text{EO}}$ is the full travelling-wave perturbation and Δn_{EO} is the amplitude of the change in the refractive index which can be evaluated using the second-order optical non-linearity (r_{ijk}) [S1]. We will revisit the phase matching in the next section (§S2) in detail. Presently, we will evaluate the refractive index perturbation produced by this RF field which only depends on the transverse mode profiles of the optical and RF modes. For this purpose, we calculate the change in the indicatrix as:

$$\begin{bmatrix} \Delta B_1 \\ \Delta B_2 \\ \Delta B_3 \\ \Delta B_4 \\ \Delta B_5 \\ \Delta B_6 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ r_{41} & r_{42} & r_{43} \\ r_{51} & r_{52} & r_{53} \\ r_{61} & r_{62} & r_{63} \end{bmatrix} \begin{bmatrix} \mathcal{E}_1^{\text{RF}}(\mathbf{r}_\perp) \\ \mathcal{E}_2^{\text{RF}}(\mathbf{r}_\perp) \\ \mathcal{E}_3^{\text{RF}}(\mathbf{r}_\perp) \end{bmatrix} \text{ and } [\mathcal{E}^{\text{RF}}(\mathbf{r}_\perp)] = \begin{bmatrix} \mathcal{E}_1^{\text{RF}}(\mathbf{r}_\perp) \\ \mathcal{E}_2^{\text{RF}}(\mathbf{r}_\perp) \\ \mathcal{E}_3^{\text{RF}}(\mathbf{r}_\perp) \end{bmatrix} \quad (\text{S7})$$

Where ΔB_i is the change in the indicatrix, $\mathcal{E}_i^{\text{RF}}(\mathbf{r}_\perp)$ is the transverse RF field and r_{lk} is the electro-optic tensor. In equation (S7), we used Neumann's principle to convert r_{ijk} to r_{lk} as described in Table S1.

Table S1: Contracted notation by using Neumann's principle [S1]

ij	11	22	33	23,32	31,13	12,21
1	1	2	3	4	5	6

We can now calculate the refractive index change by the first order perturbation theory [S1, S2],

$$\Delta n_{\text{EO}} = -\frac{\epsilon_0 n^5}{2} \frac{\int [\mathcal{E}^{\text{TE}}(\mathbf{r}_\perp)]^H \begin{bmatrix} \Delta B_1 & \Delta B_6 & \Delta B_5 \\ \Delta B_6 & \Delta B_2 & \Delta B_4 \\ \Delta B_5 & \Delta B_4 & \Delta B_3 \end{bmatrix} [\mathcal{E}^{\text{TM}}(\mathbf{r}_\perp)] d\mathbf{r}_\perp + c.c.}{\int (\mathcal{E}^{\text{TE}}(\mathbf{r}_\perp) + \mathcal{E}^{\text{TM}}(\mathbf{r}_\perp)) \cdot (\mathcal{D}^{\text{TE}}(\mathbf{r}_\perp) + \mathcal{D}^{\text{TM}}(\mathbf{r}_\perp)) d\mathbf{r}_\perp} \quad (\text{S8})$$

Here, n is the refractive index of the material, and H is the Hermitian operator. Also,

$$[\mathcal{E}^{\text{TE}}(\mathbf{r}_\perp)] = \begin{bmatrix} \mathcal{E}_1^{\text{TE}}(\mathbf{r}_\perp) \\ \mathcal{E}_2^{\text{TE}}(\mathbf{r}_\perp) \\ \mathcal{E}_3^{\text{TE}}(\mathbf{r}_\perp) \end{bmatrix} \text{ and } [\mathcal{E}^{\text{TM}}(\mathbf{r}_\perp)] = \begin{bmatrix} \mathcal{E}_1^{\text{TM}}(\mathbf{r}_\perp) \\ \mathcal{E}_2^{\text{TM}}(\mathbf{r}_\perp) \\ \mathcal{E}_3^{\text{TM}}(\mathbf{r}_\perp) \end{bmatrix} \quad (\text{S9})$$

Here we see that Δn_{EO} is a function of the selection rules between the photon's initial and final state (which depends on the transverse optical modal shape and the change in the indicatrix). We can understand this by using a simplified overlap integral which can be derived considering equation (S8). For instance,

the first component of the integral in (S10) is:

$$\begin{aligned}
& (\Delta B_1 \mathcal{E}_1^{\text{TM}}(\mathbf{r}_\perp) + \Delta B_6 \mathcal{E}_2^{\text{TM}}(\mathbf{r}_\perp) + \Delta B_5 \mathcal{E}_3^{\text{TM}}(\mathbf{r}_\perp)) \mathcal{E}_1^{\text{TE}}(\mathbf{r}_\perp) = \\
& \mathcal{E}_1^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_1^{\text{RF}} r_{11} + \mathcal{E}_1^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_2^{\text{RF}} r_{12} + \mathcal{E}_1^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_3^{\text{RF}} r_{13} \\
& + \mathcal{E}_2^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_1^{\text{RF}} r_{61} + \mathcal{E}_2^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_2^{\text{RF}} r_{62} + \mathcal{E}_2^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_3^{\text{RF}} r_{63} \\
& + \mathcal{E}_3^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_1^{\text{RF}} r_{51} + \mathcal{E}_3^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_2^{\text{RF}} r_{52} + \mathcal{E}_3^{\text{TM}} \mathcal{E}_1^{\text{TE}} \mathcal{E}_3^{\text{RF}} r_{53}
\end{aligned} \tag{S10}$$

By extension from equation (S10), we can now simplify equation (S8) into a more compact form:

$$\Delta n_{\text{EO}} \propto \int \sum_{ijk} \mathcal{E}_i^{\text{TM}}(\mathbf{r}_\perp) \mathcal{E}_j^{\text{TE}}(\mathbf{r}_\perp) r_{ijk} \mathcal{E}_k^{\text{RF}}(\mathbf{r}_\perp) d\mathbf{r}_\perp \tag{S11}$$

as written in the main text. From this equation (S11), we can easily see that to create coupling between two orthogonal states (such as TE and TM modes), we need non-zero r_{ij} with $i > 4$ to produce a change in the indicatrix for the off diagonal elements in equation (S8). Finally, we can express the electro-optic coupling rate per modal amplitude of the RF wave as,

$$g' = \frac{\omega_0 \Delta n_{\text{EO}}}{n_g v_g} \tag{S12}$$

Here, the coupling rate g' is in the units of rad/m, ω_0 is the optical frequency (we assume that $\omega_0 = \omega_1 \approx \omega_2 \gg \Omega$), $n_g = \sqrt{n_g^{\text{TM}} n_g^{\text{TE}}}$ is the geometric mean of the group indices of the modes, and $v_g = \sqrt{v_g^{\text{TM}} v_g^{\text{TE}}}$ is the geometric mean of the group velocities.

S2 Phase-matching considerations and predicted spectral characteristics

The spatial evolution of the three-wave mixing process (i.e. the interaction between the optical fields and the RF field) is described by a well-known set of coupled mode equations [S3, S4]. Presently, we assume that the process is only phase-matched for anti-Stokes scattering (see §S1) from the input optical probe (carrier) signal a_1 to the anti-Stokes signal a_2 , and an RF field b inducing the interaction. For this derivation, we assume that the RF excitation amplitude is uniform throughout the waveguide, the carrier signal is not depleted, and the optical fields do not interfere with the RF electrodes, i.e. that the beat signal of the optical waves does not excite any RF wave. Then, the governing equation for the anti-Stokes scattered signal can be written as:

$$\frac{\partial a_2}{\partial z} = ik_2 a_2 - ig' a_{1,0} b_0 e^{i(k_1+q)z} \tag{S13}$$

Here, k_1 and k_2 are the wavevectors of the carrier and scattered signals, and q represents the momentum of the RF excitation. Due to our assumptions, the carrier signal and the RF signal have the form $a_1(z) = a_{1,0} e^{ik_1 z}$ and $b(z) = b_0 e^{iqz}$ respectively where $a_{1,0}$ and b_0 are the corresponding amplitudes. Equation (S13) can be solved by assuming a solution of the form $a_2(z) = \bar{a}_2(z) e^{ik_2 z}$ where $\bar{a}_2(z)$ represents the spatially varying amplitude of the anti-Stokes signal, and the exponential term represents the propagation. Solving for $\bar{a}_2(z)$ we get:

$$\frac{\partial \bar{a}_2}{\partial z} = \left(\frac{\partial a_2}{\partial z} - ik_2 a_2 \right) e^{-ik_2 z} = -ig_{eo} a_{1,0} e^{i(k_1 + q - k_2)z} \quad (\text{S14})$$

where we define electro-optic coupling rate as $g_{eo} = g' b_0$. Then, integrating the both sides of equation (S14), we find evolution of the anti-Stokes scattered signal at the output at position $\mathcal{L}$, i.e. $\bar{a}_2(\mathcal{L})$ as:

$$\left| \frac{\bar{a}_2(\mathcal{L})}{a_{1,0}} \right|^2 = (g_{eo} \mathcal{L})^2 \text{sinc}^2((k_1 + q - k_2) \mathcal{L} / 2) \quad (\text{S15})$$

We see that the overall response follows a sinc^2 behaviour depending on the total interaction length ($\mathcal{L}$), and the phase matching condition ($k_1 + q - k_2$). The scattered power increases as the interaction length increases, and it is maximized for the perfectly phase-matched condition ($k_1 + q - k_2 = 0$). Notably, this phase matching condition is only perfectly satisfied at a one optical frequency ω_f (see Fig. S1) associated with the individual optical dispersions where the following relation is met:

$$q = k_2(\omega_f + \Omega) - k_1(\omega_f) \quad (\text{S16})$$

For any other probe frequency ω we can write the momentum mismatch as (see Fig. S1),

$$\Delta k = k_2(\omega + \Omega) - k_1(\omega) \quad (\text{S17})$$

Now the phase matching condition can be re-written to incorporate the dispersive nature of the process as:

$$\Delta k - q = \Delta q = (k_2(\omega + \Omega) - k_2(\omega_f + \Omega)) - (k_1(\omega) - k_1(\omega_f)) \quad (\text{S18})$$

Assuming linear dispersion, we can approximate (S18) as:

$$\Delta q \approx \frac{\partial k_2}{\partial \omega}(\omega - \omega_f) - \frac{\partial k_1}{\partial \omega}(\omega - \omega_f) = \frac{n_{g,2} - n_{g,1}}{c} \Delta \omega \quad (\text{S19})$$

Here, $n_{g,1}$ and $n_{g,2}$ are the group velocities of the two modes and $\Delta \omega$ is the detuning from the phase matched optical frequency (ω_f). As we substitute eqn. (S19) to equation (S15), we can directly observe that the modulation efficiency changes for different optical frequencies. From here we can calculate the full width half maximum of $\text{sinc}^2(x)$ function which is realized when $x \approx 0.4425\pi$ (i.e. $\Delta q \mathcal{L} / 2 \approx 0.4425\pi$ for our case). By using (S19), we can find the optical bandwidth as:

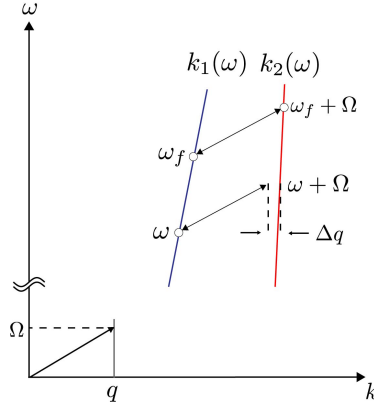


Figure S1: Phase matching condition for the anti-Stokes processes: The exact phase matching can be realized around ω_f , and for the other optical frequencies the response follows a sinc^2 function due to the group index difference.

$$\Delta\omega_{BW} \approx \frac{4 \cdot 0.4425 \cdot \pi c}{\mathcal{L}|n_{g,2} - n_{g,1}|} \quad (\text{S20})$$

Here, the additional factor of 2 is introduced since we measure the total span of the scattering bandwidth. In terms of wavelength, this bandwidth is equivalent to,

$$\Delta\lambda_{BW} \approx \frac{2 \cdot 0.4425 \cdot \lambda_0^2}{\mathcal{L}|n_{g,2} - n_{g,1}|} \quad (\text{S21})$$

For our waveguide design, we estimated the group indices for the TE (mode 2) and TM (mode 1) modes as 2.339 and 2.536 respectively by means of FEM simulations. By using equation (S21), we estimate the bandwidth for a device with length $\mathcal{L} = 1.08$ mm as $\Delta\lambda_{BW} = 2.53$ nm which is very similar to the experimental observation of ≈ 2.42 nm.

The above mode conversion analysis is based on the non-depleted pump approximation, and it can be easily be extended for larger conversion rates with coupled mode equations by additionally considering the depletion of the carrier signal. The analytical solution for this type of three-wave mixing process is well known [S5], and the scattered power for our case can be expressed as:

$$\left| \frac{\bar{a}_2(\mathcal{L})}{a_{1,0}} \right|^2 = \frac{4g_{eo}^2}{(4g_{eo}^2 + \Delta q^2)} \sin^2((4g_{eo}^2 + \Delta q^2)^{1/2} \mathcal{L}/2) \quad (\text{S22})$$

Similarly, $a_{1,0}$ is the initial carrier signal amplitude. The only difference between (S15) and (S22) is the hybridization of the optical modes under large conversion rates. The phase matching condition for the optical waves do not change and still depend on the group index difference.

S3 Calculation of non-reciprocal bandwidth

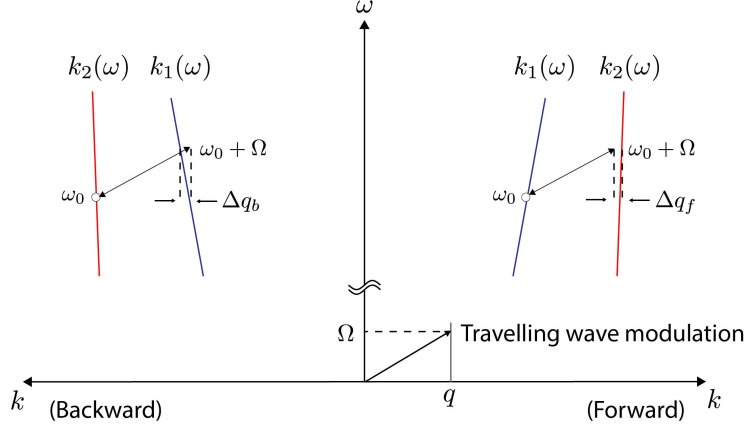


Figure S2: Phase matching conditions for the forward and backward anti-Stokes processes: The anti-Stokes scattering process for the forward and backward directions shows different requirements for phase matching condition.

In our system, non-reciprocity arises as the applied RF field breaks the time reversal symmetry [S4]. To explore this concept, we again consider the anti-Stokes processes for the forward and backward directions at some optical probing frequency ω_0 (similar equations can be derived for the Stokes process) as shown in Fig. S2. For the forward direction, for a probe injected into mode 1, the phase mismatch is given by

$$\Delta q_f = k_2(\omega_0 + \Omega) - k_1(\omega_0) - q . \quad (\text{S23})$$

Similarly, in the backward direction for a probe injected to mode 2, the phase mismatch is,

$$\Delta q_b = k_2(\omega_0) - k_1(\omega_0 + \Omega) - q \quad (\text{S24})$$

Here, the probing mode in the backward direction is different since the forward directed momentum of the applied modulation never allows for anti-Stokes scattering from mode 1 to mode 2 in this direction. Then, taking difference between these two equations, we find that,

$$\Delta q_f - \Delta q_b = (k_2(\omega_0 + \Omega) - k_2(\omega_0)) - (k_1(\omega_0) - k_1(\omega_0 + \Omega)) \quad (\text{S25})$$

If we assume linear dispersion for the frequency of interest, the above equation simplifies by using Taylor's expansion,

$$\Delta q_f - \Delta q_b \approx \frac{\Omega}{c}(n_{g,2} + n_{g,1}) \quad (\text{S26})$$

From Eqn. (S26), we see that the phase matching conditions for the forward and backward directions depend on the modulation frequency and the group index. Importantly, this shows that the anti-Stokes process is not reciprocal (similarly the Stokes process). Since the forward and backward directions require different phase matching conditions (Δq_f and Δq_b), the phase matching frequencies are also different from each other (Fig. S3).

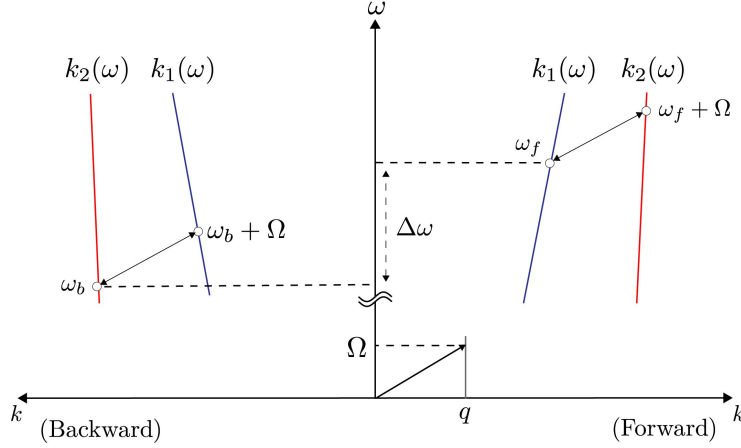


Figure S3: The spectral shift for the anti-Stokes processes: The group index difference between the optical modes results in different phase matching frequencies for the forward and backward directions.

The difference between ω_f and ω_b can be found by considering the perfect phase matching condition for the anti-Stokes process in the backward direction. When this occurs (i.e. $\Delta q_b = 0$), the forward anti-Stokes process accumulates some phase ($\Delta q_f \neq 0$) which can be predicted by using equation (S26). Similarly, the same phase accumulation can be calculated by equation (S19) which identifies the optical frequency ω_f where the phase matching will be perfect, i.e. $\Delta q = \Delta q_f$. Thus by using equations (S19) and (S26) we can calculate the frequency separation as:

$$(\omega_b - \omega_f) \frac{n_{g,1} + n_{g,2}}{c} = \Omega \frac{n_{g,2} - n_{g,1}}{c} \quad (\text{S27})$$

$$\omega_b - \omega_f = \frac{n_{g,1} + n_{g,2}}{n_{g,2} - n_{g,1}} \Omega \quad (\text{S28})$$

For significant non-reciprocity, this spectral shift (S28) should be larger than modulation bandwidth (S20), for which we require,

$$\frac{n_{g,1} + n_{g,2}}{|n_{g,2} - n_{g,1}|} \Omega \gg \frac{4 \cdot 0.4425 \cdot \pi c}{\mathcal{L} |n_{g,2} - n_{g,1}|} \quad (\text{S29})$$

which is equivalent to

$$\mathcal{L} \gg \frac{4 \cdot 0.4425 \cdot \pi c}{(n_{g,1} + n_{g,2})\Omega} \quad (\text{S30})$$

As can be seen from eqn.(S30), the nonreciprocal contrast enhances with larger values of $\mathcal{L}$ or larger values of Ω . In practice, we limit $\mathcal{L}$ to stay at or below the length for full conversion (guided by equation S22). For our waveguide design, we estimated the group indices for the TE (mode 2) and TM (mode 1) modes as 2.339 and 2.536 respectively. By using our experiment parameters, we estimate the spectral shift of a device with length of $\mathcal{L} = 1.08$ mm and a modulation frequency of $\Omega = 6$ GHz to be to be 0.302 nm which is close to the experimental observation of ≈ 0.3 nm (See Fig. 5)

S4 Device design and generation of the synthetic traveling wave excitation

The first task in designing this device is to select TE and TM modes with appropriate symmetries and a sufficiently small momentum gap so that practical electrode geometries can be realized. We need to arrive at these decisions iteratively given the relationships discussed below.

Our next goal is to generate a traveling wave modulation corresponding to the required momentum q that bridges the gap between the TE and TM modes. We aim to achieve this using integer N samples per period of this wave with each sample corresponding to the voltage on one electrode. A lower bound constraint on N is easy to establish from the Nyquist sampling theorem. For $N = 2$ the corresponding discrete frequency is π , which is exactly at the Nyquist limit and corresponds to a standing wave excitation without any identifiable directionality. For $N \geq 3$ the spatial sampling rate is past the Nyquist limit and the wave directionality becomes well defined. Therefore $N = 3$ is the minimum number of sampling points per period if we constrain N to be an integer. This same argument has been leveraged in previous works [S6–S8].

In experiments we generated the excitation using a split electrode structure (as sketched in Fig. S4a). Mathematically speaking, the split electrode structure represents a spatial sample-and-hold operation of the underlying sinusoidal wave $g(z, t) = \frac{1}{2}(e^{i(qz - \Omega t)} + e^{-i(qz - \Omega t)})$ (as shown in Fig. S4b), or alternatively a space-domain convolution of samples of the underlying sinusoid with an appropriately scaled $\text{rect}(z/\mathfrak{L})$ function where $\mathfrak{L} \leq L$ is the physical length of each electrode (L is the pitch of the electrodes). This operation produces additional terms in the momentum spectrum that appear at integer multiples of q . We can explicitly calculate this momentum spectrum by mathematically describing the sampling process. First, our true stimulus function $g_s(z)$ can be expressed as:

$$g_s(z, t) = \text{rect}\left(\frac{z}{\mathfrak{L}}\right) \circledast \sum_{\forall n \in \mathbb{Z}} g(z, t) \delta(z - nL) \quad (\text{S31})$$

Where ‘ $\circledast$ ’ represents the convolution operation, L is the electrode pitch which identifies the sampling period. The δ -function extracts samples (in space z)

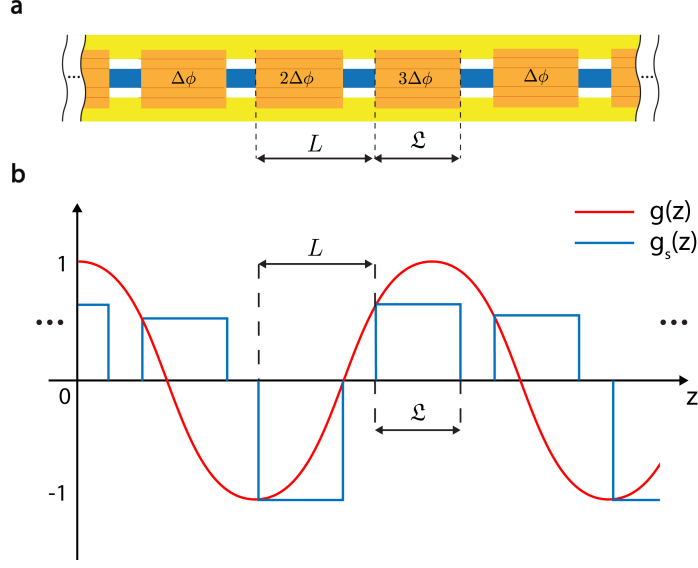


Figure S4: Set up for a synthetic traveling wave with spatial sampling. (a) The split electrode structure within the device, showing key spatial parameters and the relative phase shifts using 3 electrodes per period. (b) A visual representation of the underlying sinusoidal wave $g(z, t)$ that we emulate using split electrodes. The actual spatially sampled function that is generated by the electrodes is represented by $g_s(z, t)$. Both waves are plotted at a time instant.

from the underlying $g(z, t)$. We can now take a Fourier transform (mapping space z to momentum k) to obtain the spectrum:

$$G_s(k, t) = \frac{\sin(k\mathcal{L}/2)}{k\mathcal{L}/2} \frac{\pi\mathcal{L}}{L} \sum_{\forall n \in \mathbb{Z}} \left[\delta\left(k - q - \frac{2\pi n}{L}\right) e^{-i\Omega t} + \delta\left(k + q - \frac{2\pi n}{L}\right) e^{+i\Omega t} \right] \quad (\text{S32})$$

Recall that our design choice of N and requirement of momentum q sets an electrode pitch constraint of $L = 2\pi/Nq$. Considering this, we can rewrite Eqn. S32 and further take the Fourier transform in time:

$$G_s(k, \Omega) = \frac{\sin(k\mathcal{L}/2)}{k\mathcal{L}/2} \frac{2\pi^2\mathcal{L}}{L} \sum_{\forall n \in \mathbb{Z}} [\delta(k - (nN + 1)q) \delta(\omega + \Omega) + \delta(k - (nN - 1)q) \delta(\omega - \Omega)] \quad (\text{S33})$$

From Eqn. S33, we see that the synthetic wave spectrum only has components at $(nN \pm 1)q$ with modulation frequency $\pm\Omega$. Here, $N \geq 3$ is fixed due to the directionality constraint described above, and n is any integer. These spectral

components are additionally scaled by a sinc envelope implying that the $n = 0$ fundamental components at $\pm q$ will have the largest amplitude. Finally we note that, pairwise, one momentum component is associated with a frequency up-shift while the other is associated with a down-shift, confirming the broken time-reversal symmetry.

At this point it is also interesting to consider the limiting case where $N \rightarrow \infty$, i.e. the continuum case. This forces both $L \rightarrow 0$ and $\mathfrak{L} \rightarrow 0$, and the pre-factor $\sin(k\mathfrak{L}/2)/(k\mathfrak{L}/2) \times (\mathfrak{L}/L)$ then approaches 1 in the limit. It can then be shown that the resulting spectrum is then only composed of momentum components at $\pm q$, i.e. a pure sinusoid.

As an example closely related to our paper, in Fig. S5 we plot the momentum spectrum for a system using three phase excitation (i.e., $N = 3$) and sampling interval L . Here we assume that $\mathfrak{L} = L$ for simplicity. We see that the corresponding fundamental momentum components are $k = \pm 2\pi/3L$, and are largest in amplitude. For the experiment, the choice of N sets the RF phase gradient between adjacent electrodes to $\Delta\phi = \pm 2\pi/N$ to ensure periodicity. The choice of sign for $\Delta\phi$ sets the direction of the propagating wave.

In practice, the electrical parasitic capacitance between electrodes needs to be considered. If this parasitic impedance is small (due to large capacitance), the applied RF signal would simply be shunted and it will be difficult to develop the required voltage on the electrodes. This will hinder the performance of the device. These parasitic capacitances can be large if the electrodes are too close together, i.e. if N is too large or if $\mathfrak{L}$ is too close to L . In our experiments we set $L = 120\mu m$ and $\mathfrak{L} = 100\mu m$ to keep a $20\mu m$ gap between adjacent electrodes.

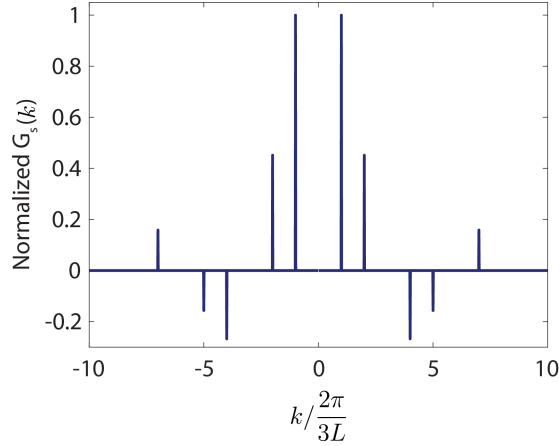


Figure S5: Example momentum spectrum of the synthetic wave corresponding to our experiments. $N = 3$ with a sampling interval L , which is equal to $\mathfrak{L}$. The fundamental momentum components are $k = \pm 2\pi/3L$.

S5 Electrical analysis of the electrodes and cut-off frequency

For our split electrode design, we must ensure that the RF excitation is in the standing wave regime as it provides the flexibility to engineer the synthetic momentum with lithographically defined features (electrode pitch and sampling). In contrast, the propagating wave regime [S9] is useful for phase modulators to satisfy the momentum matching at very high frequencies [S9–S11] (above 100 GHz). In the standing wave regime the electrode will act as a simple RLC circuit [S12] (See Fig. S6a). It is important for us to understand the cut-off frequency over which the standing wave behavior will be maintained. Beyond this frequency, the transferred voltage to the modulating capacitor reduces, leading to inefficient electro-optic modulation.

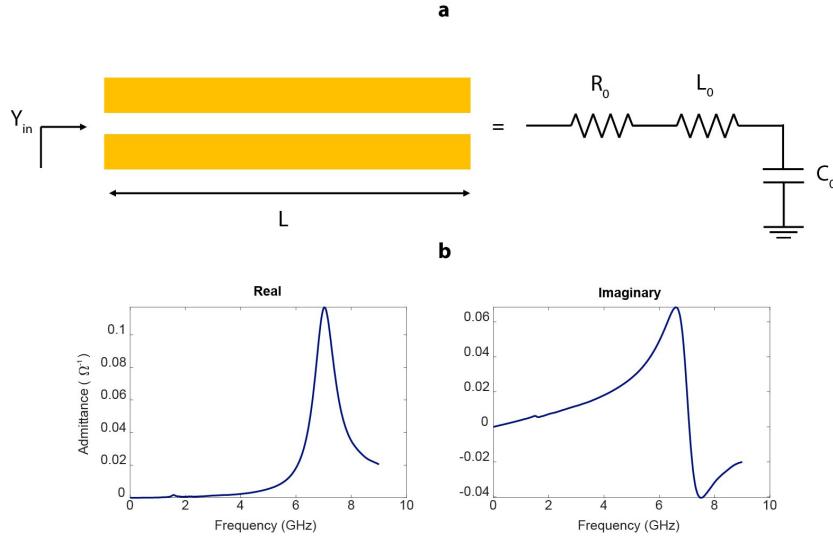


Figure S6: Electrical modelling of the RF electrodes (a) Circuit equivalent of the electrodes for standing wave region [S12]. (b) Admittance measurements of the electrodes by a vector network analyzer.

We first measure the Y parameters (admittance parameters) for our electrodes. As the inductive component is not significant at low frequencies, we can estimate the resistivity and capacitance of our structure with measurements at 10 kHz. Here, we estimate $C_0 = 0.6$ pF and $R_0 = 9$ Ω . The capacitance value agrees well with the finite element simulations, however the resistance is larger than the expected value of 3 Ω . This is likely related to the quality of the gold film used in the electrodes since it is deposited by e-beam evaporation. We then estimated the inductance value as $L_0 = 0.83$ nH by using the standing wave resonance of the structure, as can be seen in the measurements shown in Fig. S6b. With these parameters, we can confirm that the cut-off

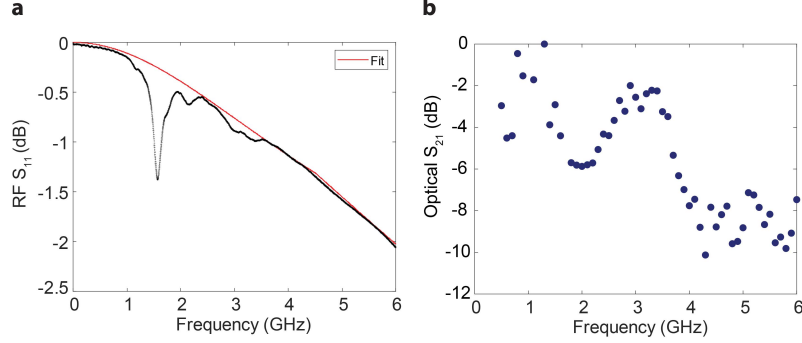


Figure S7: Modulation response (a) The reflection coefficient (S_{11}) of the electrodes is measured by a vector network analyzer demonstrating an RC response ensuring standing wave approximation. (b) Optical S_{21} parameter for different optical frequencies

frequency for this transmission line is relatively large (>10 GHz) and that at our operational frequency of 0-6 GHz the standing wave approximation is valid. We additionally present the RF S_{11} measurement in Fig. S7a, which displays an RC response over the operational frequency range as expected. The little dip around 1.5 GHz frequency is due to a resonance in the cables caused due to back-reflection from the electrode. After measuring the electrical parameters, we additionally performed an optical S parameter analysis to understand the modulation response for different frequencies. The results of this measurement is also given in Fig. S7b.

In experiments, the RF signal generator assumes a $50\ \Omega$ load and sets the output voltage (i.e., sinusoid amplitude) accordingly. If we set the signal generator output power as P_{drive} , the voltage across the signal generator terminals can be calculated by considering the power dissipation at the assumed $50\ \Omega$ load (i.e. $P_{drive} = (1/2) \times V_{load} \times I_{load} = V_{source}^2 / (50 \times 4 \times 2)$). As an example, if we set the RF drive power at 21 dBm, then the sinusoid amplitude at the signal generator is $V_{source} = \sqrt{2 \times 4 \times 50 \times P_{drive}} \approx 7.1V$. Since the device electrodes form a capacitor, they appear like an open circuit, causing all of this voltage to appear on the electrodes. Alongside, almost all the input RF power is reflected back to the signal generator due to the open circuit condition at the capacitive electrodes.

S6 Important cross-sectional dimensions

For our experimental demonstration, we optimized the electrode distance in order to maximize the electro-optic coupling rate while minimizing the optical loss. For our design, the bottom electrode distance to our optical structure is chosen to be $1.75\ \mu m$ for a waveguide width of $0.5\ \mu m$. The cladding thickness (and hence the top electrode distance) is chosen to be $1.5\ \mu m$ since the optical

fields are well confined in the vertical direction. All other dimensions of our structure are presented in Fig.S8.

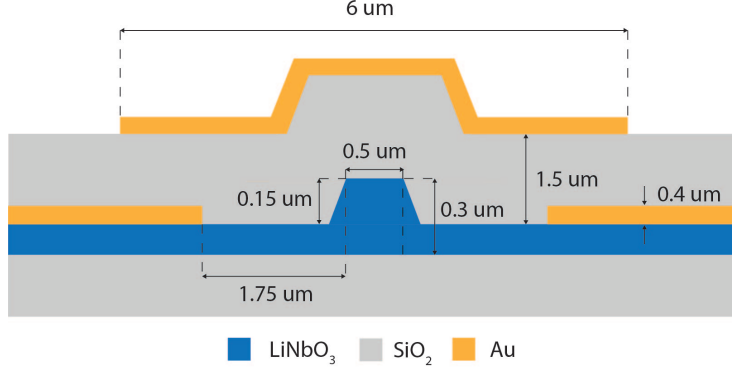


Figure S8: Physical dimensions of the EO-NPR waveguide cross-section: The dimension for the structure are chosen to maximize electro-optic coupling rate while minimizing the optical loss rate.

S7 Optical characterization

In order to measure the Stokes, anti-Stokes, and the carrier transmission, we utilize an optical heterodyne detection system by using an acousto-optic frequency shifter (Fig. S9)

For our experiment as given in the Main Manuscript Fig. 3, we first measure TE to TE and TM to TM transmissions through the device and gratings without any RF stimulus applied. For the TE case, we can write the optical field arriving at the photo-detector as;

$$s_{\text{TE}} = \frac{s_{\text{in}} e^{i\omega_l t} \sqrt{G_{\text{TE}}} \sqrt{G_{\text{TE}}}}{\sqrt{2}} + s_r e^{i(\omega_l + \Omega_r)t} \quad (\text{S34})$$

Here, the first term represents the TE to TE transmission, and the second term represents the modulated optical signal due to AOFS. G_{TE} is the power transmission coefficient for the TE gratings, and $\sqrt{2}$ appears due to our Y-splitter near the output. Similarly, for TM case, the the optical field becomes;

$$s_{\text{TM}} = \frac{s_{\text{in}} e^{i\omega_l t} \sqrt{G_{\text{TM}}} \sqrt{G_{\text{TM}}}}{\sqrt{2}} + s_r e^{i(\omega_l + \Omega_r)t} \quad (\text{S35})$$

Similarly, G_{TM} is the power transmission of the gratings. The resulting RF signal from the photodetector is the beat note at Ω_r and can be expressed for these two cases as;

$$P_{\Omega_r, \text{TE}} = g_{pd} |s_r|^2 \frac{|s_{\text{in}}|^2 G_{\text{TE}}^2}{2} \quad (\text{S36})$$

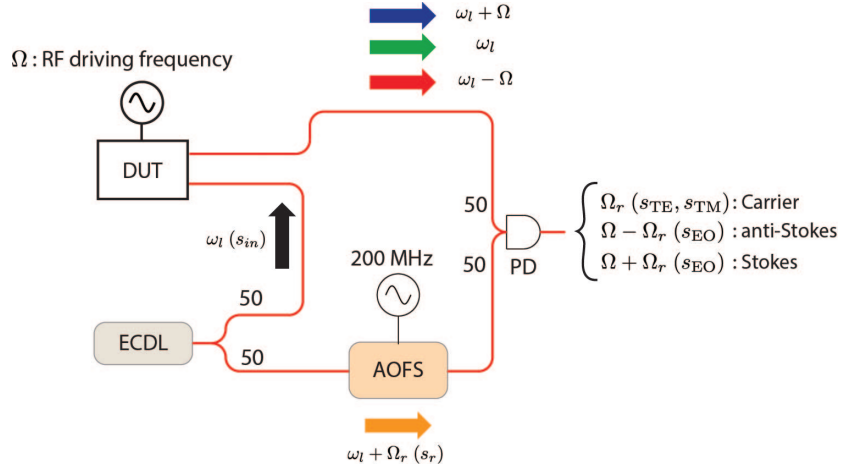


Figure S9: Heterodyne detection system ECDL: External cavity diode laser, AOFS: Acousto-optic frequency shifter, PD: Photodetector, DUT: Device under test, Ω_r : Operating frequency of the AOFS (i.e. 200 MHz)

$$P_{\Omega_r, \text{TM}} = g_{pd} |s_r|^2 \frac{|s_{in}|^2 G_{\text{TM}}^2}{2} \quad (\text{S37})$$

Here, g_{pd} is the sensitivity of the photo-detector. We plot these RF power measurements in Fig. S10.

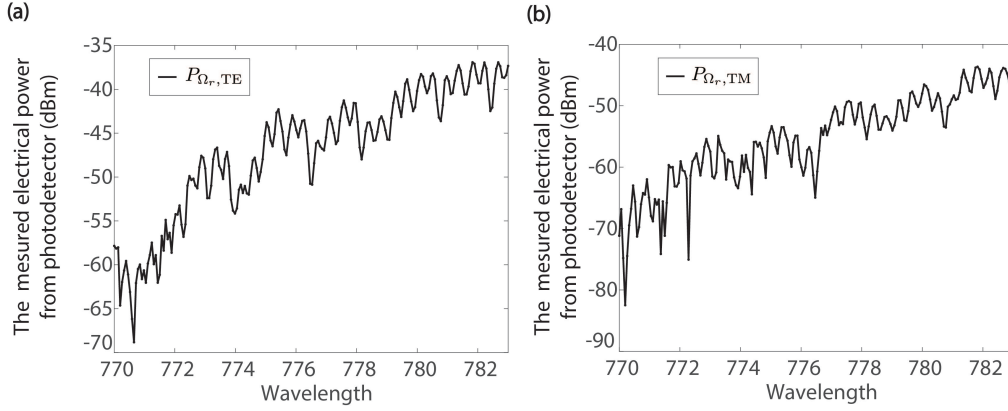


Figure S10: Optical characterization (a) The measured electrical signal for TE to TE transmission for the wavelength of interest (770 nm - 783 nm). (b) The measured electrical signal for TM to TM transmission for the wavelength of interest (770 nm - 783 nm).

We then perform the inter-polarization scattering experiment (Main Manuscript Fig. 4-5) where we inject TM mode and receive TE sidebands. For this case,

the measured optical spectrum at the photo-detector can be expressed as;

$$s_{EO} = g_{eo} \mathcal{L} \text{sinc}(\Delta q_S \mathcal{L}/2) \frac{s_{in} e^{i(\omega_l - \Omega)t} \sqrt{G_{TE}} \sqrt{G_{TM}}}{\sqrt{2}} + g_{eo} \mathcal{L} \text{sinc}(\Delta q_{AS} \mathcal{L}/2) \frac{s_{in} e^{i(\omega_l + \Omega)t} \sqrt{G_{TE}} \sqrt{G_{TM}}}{\sqrt{2}} + s_r e^{i(\omega_l + \Omega_r)t} \quad (\text{S38})$$

Here, Δq_S and Δq_{AS} are the phase mismatch for the Stokes and anti-Stokes processes. The resulting RF electrical signals from the photodetector are the beat notes of the optical reference signal with the sidebands which occur at $\Omega + \Omega_r$ and $\Omega - \Omega_r$. We can express these signals as;

$$P_{\Omega + \Omega_r} = g_{pd} |s_r|^2 (g_{eo} \mathcal{L} \text{sinc}(\Delta q_S \mathcal{L}/2))^2 \frac{|s_{in}|^2 G_{TE} G_{TM}}{2} \quad (\text{S39})$$

$$P_{\Omega - \Omega_r} = g_{pd} |s_r|^2 (g_{eo} \mathcal{L} \text{sinc}(\Delta q_{AS} \mathcal{L}/2))^2 \frac{|s_{in}|^2 G_{TE} G_{TM}}{2} \quad (\text{S40})$$

Our goal is to determine the sideband power relative to the carrier, without considering the grating efficiencies or the Y-splitter. This can be accomplished using Eqns. (S36) and (S37) by taking the ratio:

$$\bar{P}_{\Omega + \Omega_r} = \frac{P_{\Omega + \Omega_r}}{\sqrt{P_{\Omega_r, TE} P_{\Omega_r, TM}}} = (g_{eo} \mathcal{L} \text{sinc}(\Delta q_S \mathcal{L}/2))^2 \quad (\text{S41})$$

$$\bar{P}_{\Omega - \Omega_r} = \frac{P_{\Omega - \Omega_r}}{\sqrt{P_{\Omega_r, TE} P_{\Omega_r, TM}}} = (g_{eo} \mathcal{L} \text{sinc}(\Delta q_{AS} \mathcal{L}/2))^2 \quad (\text{S42})$$

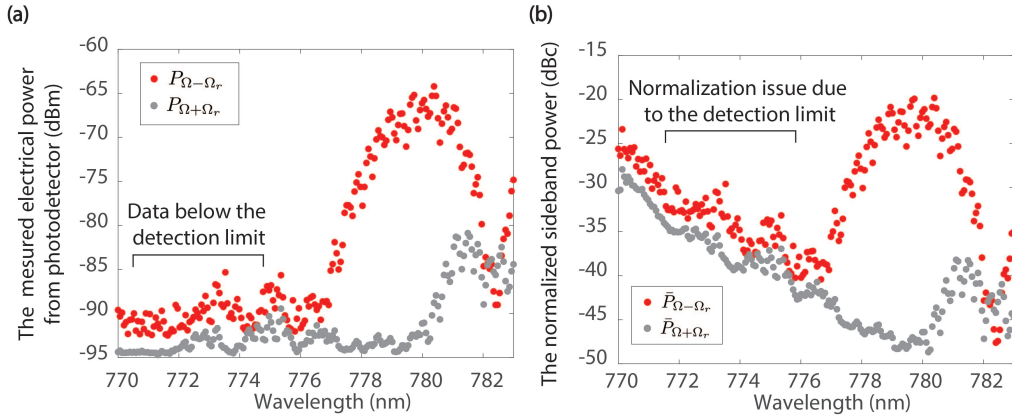


Figure S11: Sideband measurements (a) The measured electrical signal for TM to TE conversion process for the wavelength of interest (770 nm - 783 nm). (b) The normalized electrical signal for (by using equations S41 and S42) TM to TE conversion process for the wavelength of interest (770 nm - 783 nm).

On the dB scale, the equations (S41-S42) become;

$$\bar{P}_{\Omega+\Omega_r}^{\text{dB}} = P_{\Omega+\Omega_r}^{\text{dB}} - \frac{1}{2}(P_{\Omega_r, \text{TE}}^{\text{dB}} + P_{\Omega_r, \text{TM}}^{\text{dB}}) \quad (\text{S43})$$

$$\bar{P}_{\Omega-\Omega_r}^{\text{dB}} = P_{\Omega-\Omega_r}^{\text{dB}} - \frac{1}{2}(P_{\Omega_r, \text{TE}}^{\text{dB}} + P_{\Omega_r, \text{TM}}^{\text{dB}}) \quad (\text{S44})$$

In Fig. S11 we plot both the measured and the normalized electrical signals for TM to TE conversion. We see that due to the small transmission of the gratings near 770 nm, the spectrum analyzer measures noise rather than the converted signal, which causes problem during normalization. For this reason, the only reliable data is from 775 nm to 783 nm (as plotted in the main manuscript) and can be fitted with the theory. From this, we can extract the peak electro-optic modulation strength by using equations (S15, S42) and can predict performance of a much longer device via equation (S22).

By means of the experimental measurements in Fig. S11, we calculate the electro-optic conversion rate to be $g_{eo} = 94.21 \text{ rad/m}$ for our demonstrated EO-NPR with 21 dBm drive power. Through equation (S22) we can now estimate that the length required for full conversion will be 16.74 mm if the same g_{eo} is maintained along the full length. Since the applied power is 21 dBm (for each electrode) we estimate, with the support of the electrode characterization measurements, that the voltage on the modulating electrodes is $\approx 7.1 \text{ V}$ (sinusoid amplitude) allowing us to estimate the voltage-length product as $7.1 \text{ V} \times 1.67 \text{ cm} \approx 11.85 \text{ V} \cdot \text{cm}$. When we normalize the conversion rate with our applied voltage, we find $\approx 0.13 \text{ rad/V} \cdot \text{cm}$ which quantifies electro-optic polarization rotation in a manner similar to the Verdet constant in magneto-optics.

S8 Potential performance enhancements by improving electrode design

The current design of the EO-NPR uses three electrodes (in cross-section) for generating the electro-optic modulation: one on the top and two on the sides of the optical waveguide (Fig.S12a), enabling vertical fringe electric field distribution across the waveguide. To further enhance the electric field distribution and improve the polarization rotation rate, an additional fourth grounded electrode may be introduced just below the optical waveguide (Fig.S12b). This configuration focuses the vertical electric field distribution, enhances the estimated polarization rotation rate by approximately 15%.

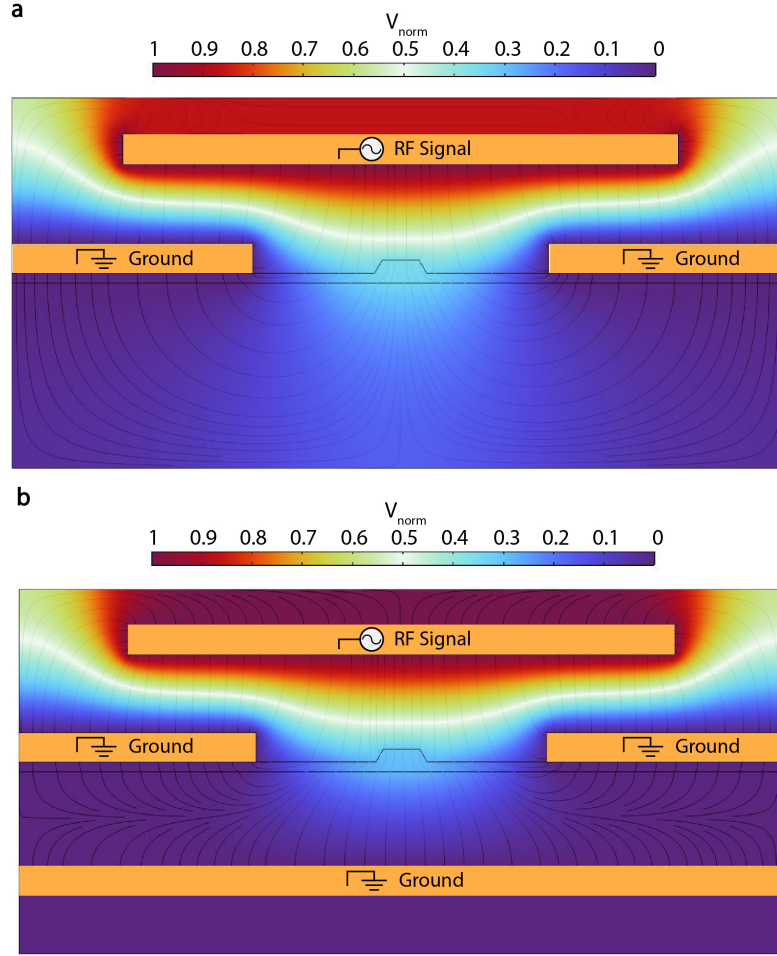


Figure S12: Comparison of the electric field distributions with different electrode designs (a) Current electrode structure: single top and two side electrodes to realize vertical electric field distribution within the optical waveguide **(b)** Improved electrode structure: incorporating a fourth electrode below the waveguide to optimize vertical electric field distribution within the optical waveguide

References

- [S1] Weis, R. S. & Gaylord, T. K. Lithium niobate: Summary of physical properties and crystal structure. *Applied Physics A* **37**, 191–203 (1985).
- [S2] Balram, K. C., Davanço, M., Lim, J. Y., Song, J. D. & Srinivasan, K. Moving boundary and photoelastic coupling in gaas optomechanical resonators. *Optica* **1**, 414–420 (2014).
- [S3] Kharel, P., Behunin, R. O., Renninger, W. H. & Rakich, P. T. Noise and dynamics in forward brillouin interactions. *Physical Review A* **93**, 063806 (2016).

- [S4] Kittlaus, E. A., Otterstrom, N. T., Kharel, P., Gertler, S. & Rakich, P. T. Non-reciprocal interband brillouin modulation. *Nature Photonics* **12**, 613–619 (2018).
- [S5] Yariv, A. Coupled-mode theory for guided-wave optics. *IEEE Journal of Quantum Electronics* **9**, 919–933 (1973).
- [S6] Sounas, D. L. & Alù, A. Angular-momentum-biased nanorings to realize magnetic-free integrated optical isolation. *ACS Photonics* **1**, 198–204 (2014).
- [S7] Tian, H. *et al.* Magnetic-free silicon nitride integrated optical isolator. *Nature Photonics* **15**, 828–836 (2021).
- [S8] Tian, H. *et al.* Magnetic-free silicon nitride integrated optical isolator. *arXiv:2104.01158* (2021).
- [S9] Ghione, G. *Semiconductor devices for high-speed optoelectronics* (Cambridge University Press, 2009).
- [S10] Wang, C. *et al.* Integrated lithium niobate electro-optic modulators operating at cmos-compatible voltages. *Nature* **562**, 101–104 (2018).
- [S11] Zhu, D. *et al.* Integrated photonics on thin-film lithium niobate. *Adv. Opt. Photon.* **13**, 242–352 (2021).
- [S12] Kim, M.-G., Lee, B. H. & Yun, T.-Y. Equivalent-circuit model for high-capacitance mlcc based on transmission-line theory. *IEEE Transactions on Components, Packaging and Manufacturing Technology* **2**, 1012–1020 (2012).